

Discovering the Origins of Gravitational-Wave Sources

Draft Thesis Review

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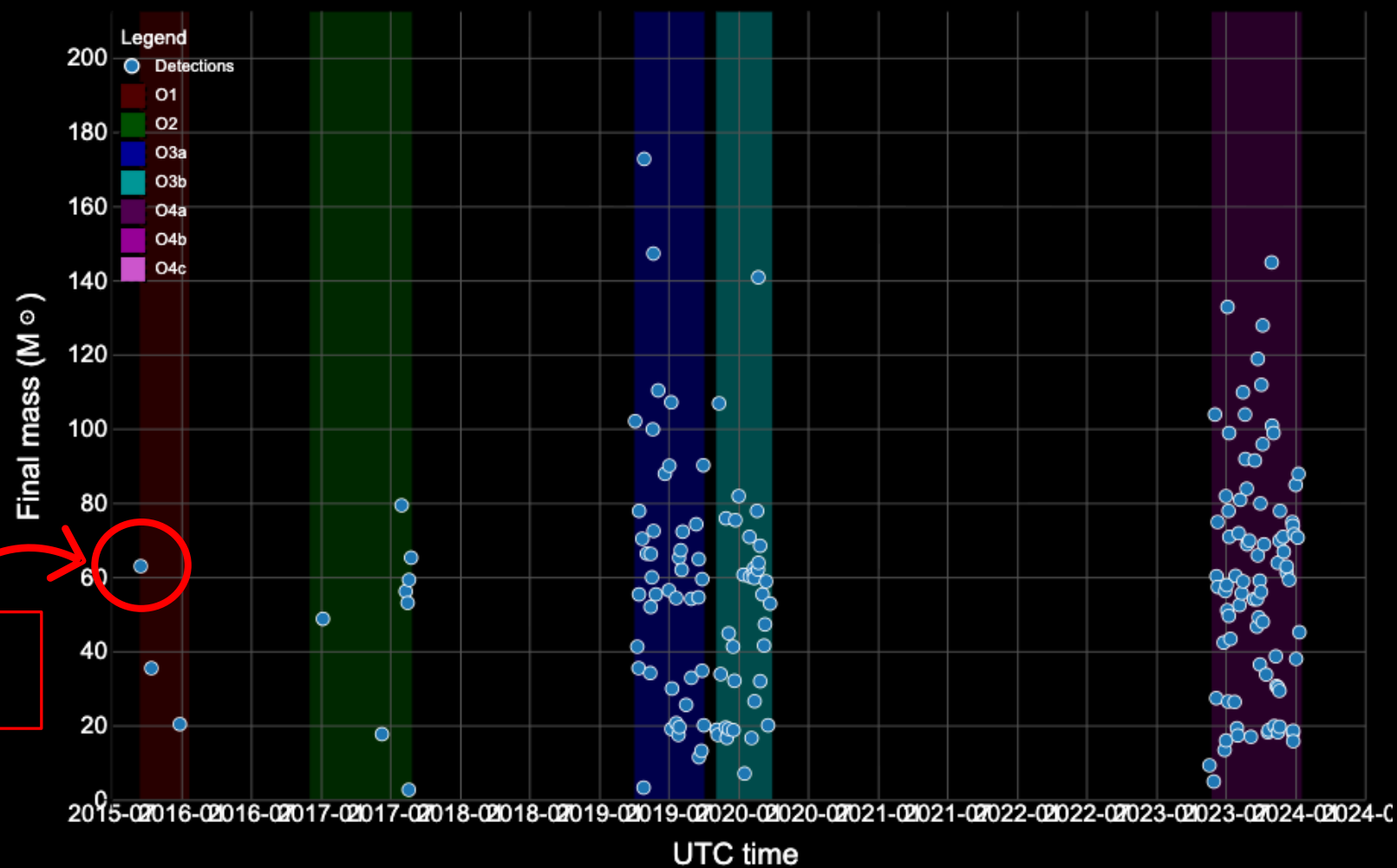


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Supercomputing



LIGO-Virgo-KAGRA Compact Binary Catalogue



Binary Black Hole
GW150914

credit: cardiffgravity.org

Why care about double neutron stars?



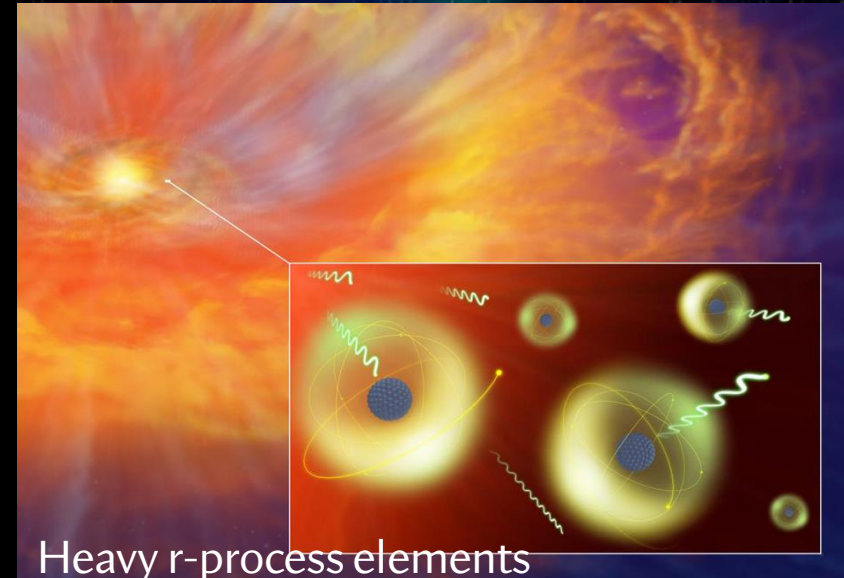
Supernovae explosion



Gamma ray bursts



Kilonovae



Heavy r-process elements

Radio Telescopes



Parkes Radio Observatory

~ 26 Galactic DNS in the field
Total mass: $2.5\text{--}2.9 M_{\odot}$

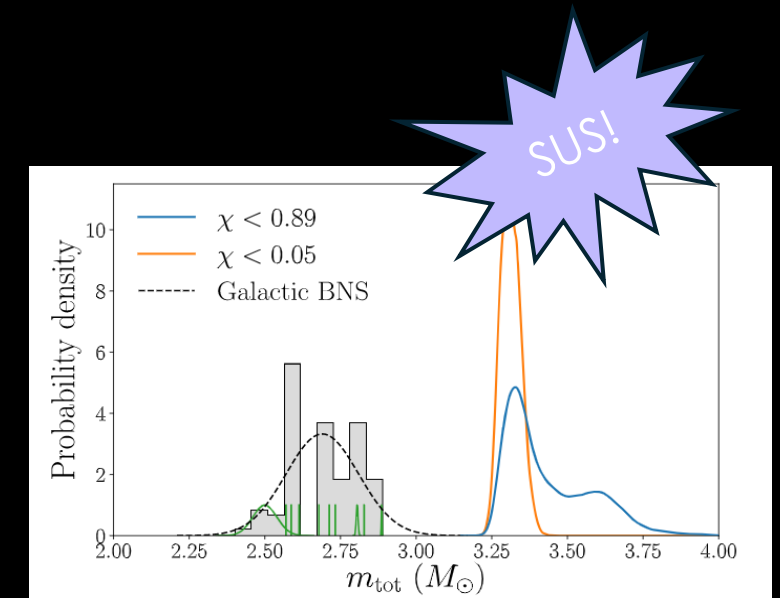
Gravitational Wave Observatories



LIGO Livingston

GW170817: Total mass : $\sim 2.8 M_{\odot}$
GW190425: Total mass : $\sim 3.4 M_{\odot}$

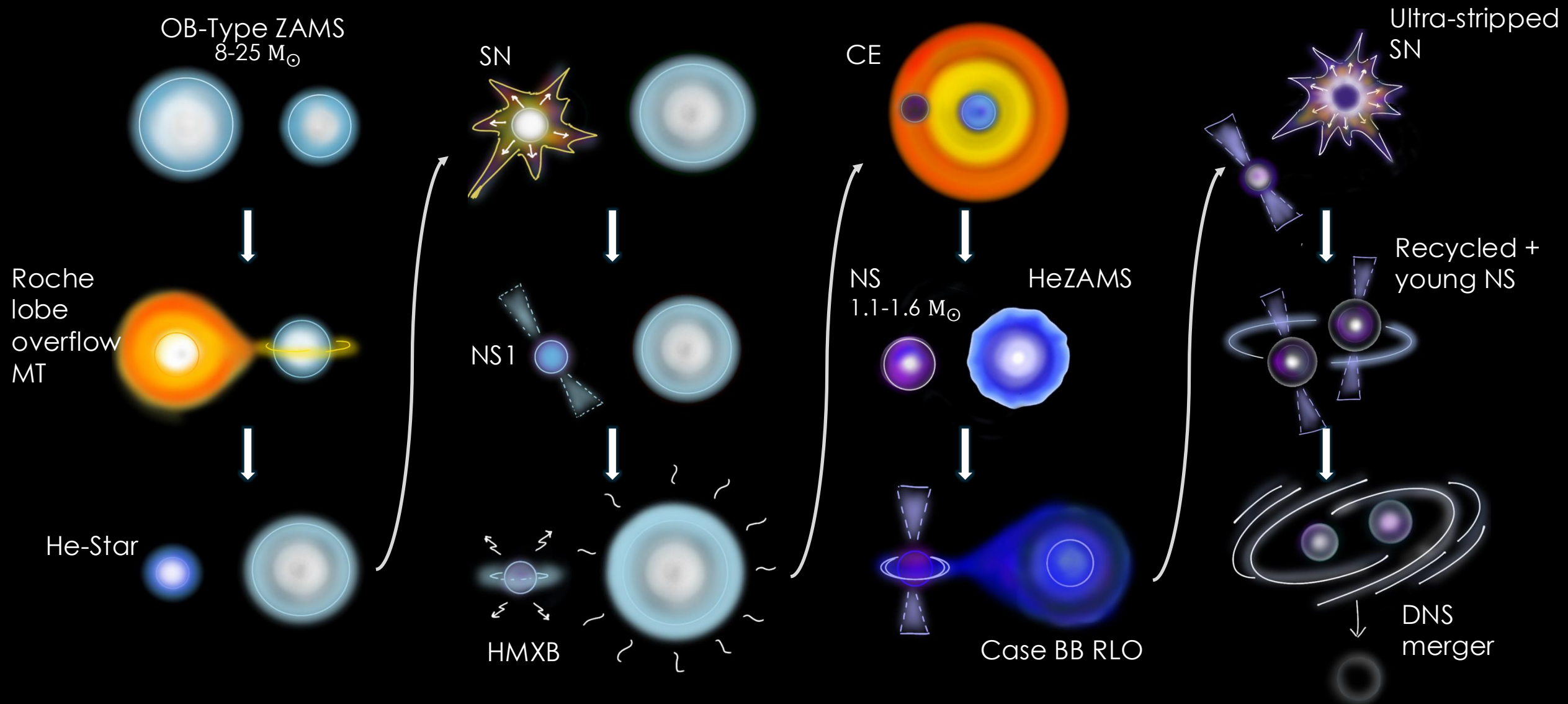
GW231109_235456 : $\sim 2.95 M_{\odot}$ (if astrophysical!)



Abbot et al. 2020

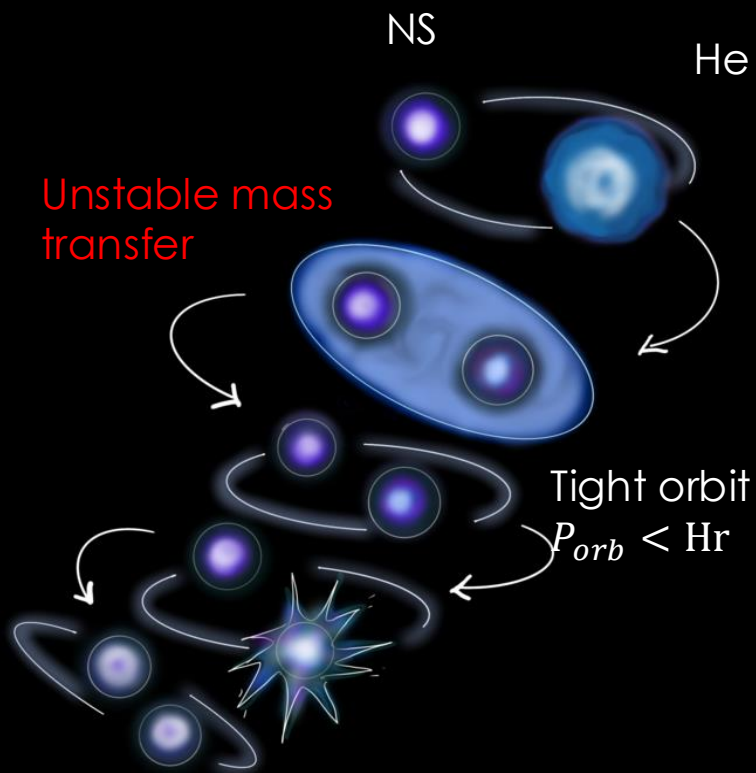
DNS Total mass $\geq 3 M_{\odot}$: Heavy DNSs

Standard Double Neutron Star Formation Pathway at $Z_{\odot}$ – T. Tauris et al. 2017



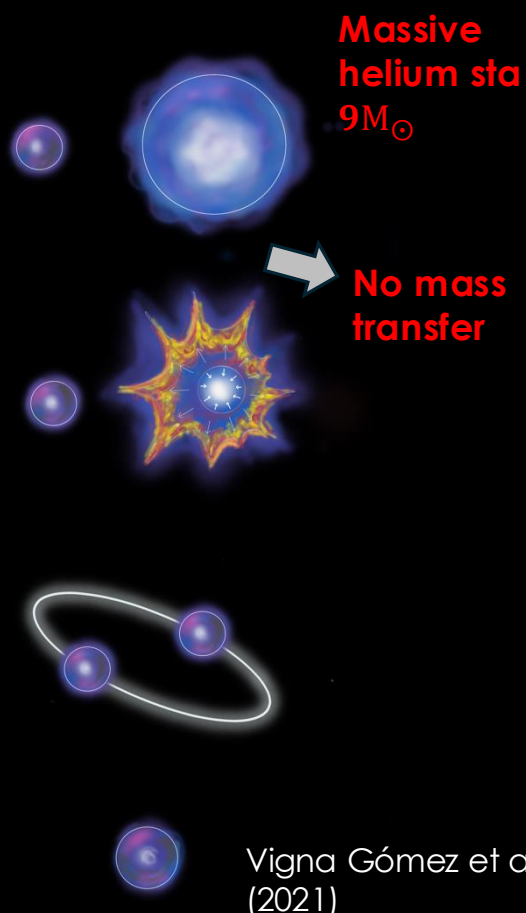
Alternate formation channels/pathways

Fast Merger

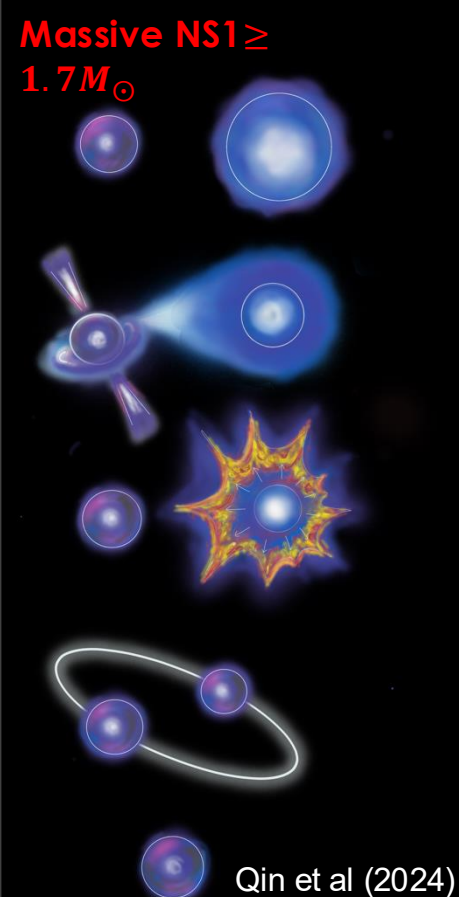


B.P. Abbott et al. (2020),
Romero

Massive He star



Massive NS1



Low metallicity

(Same as standard channel)

?

Giacobbo & Mapelli (2018)

Basically, similar formation pathway just different initial conditions and following mass transfer processes!

No self-consistent, unbiased formation pathway to explain heavy double neutron star formation!

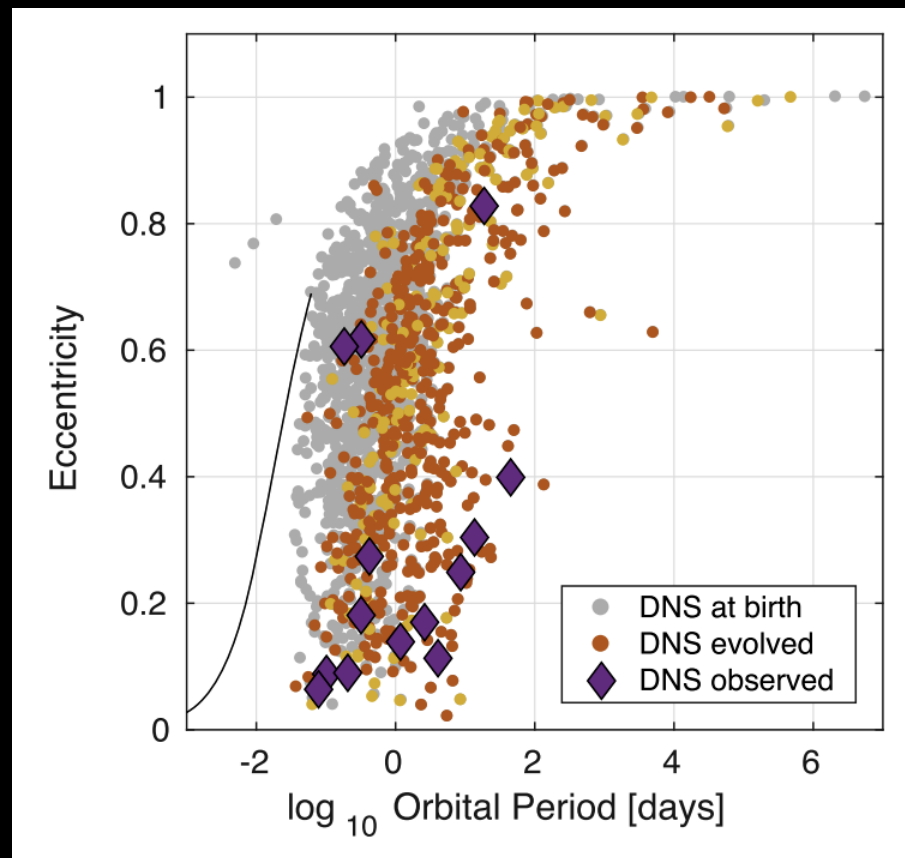
Common approach for population studies

Rapid Binary Population Synthesis

e.g., COMPAS (Stevenson et al. 2017),
COMBINE (Kruckow et al., 2018)

Caveat : Fail to reproduce the Galactic DNS properties!

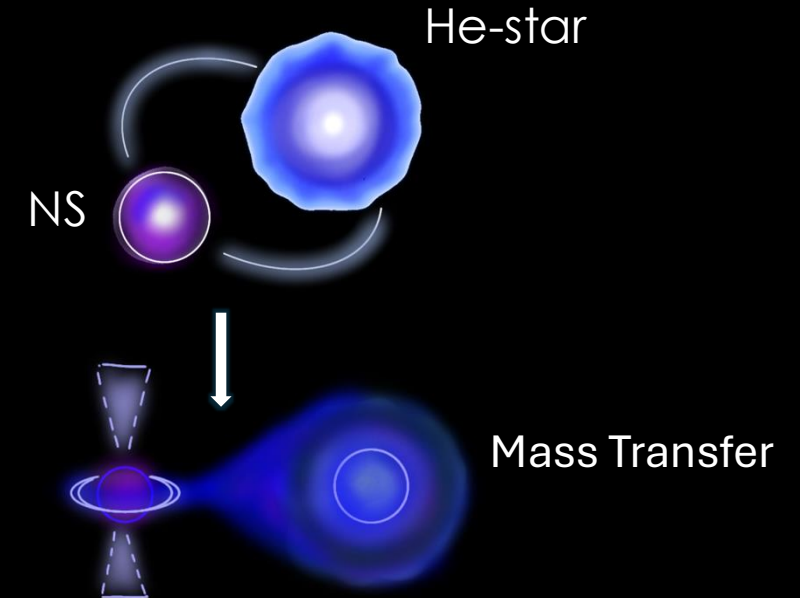
e.g., Orbital period – eccentricity distribution



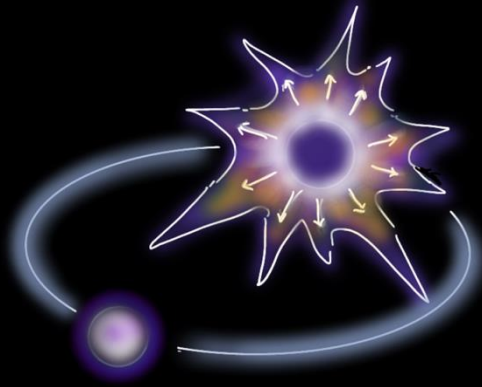
Vigna Gómez et al. (2018)

*Project I, Part A***Simulation of Double Neutron Stars**
(using detailed stellar evolution code)**Initial parameter space**He ZAMS mass: $2.5 - 10 M_{\odot}$ First-born NS (NS1) mass : $1.4 M_{\odot}$

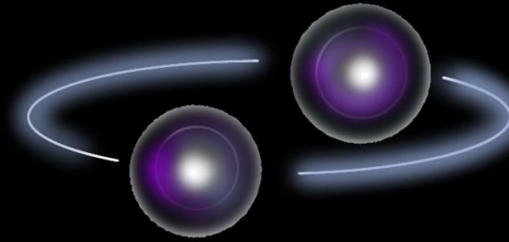
Orbital period : 1.5 hrs – 100 days

Metallicity : $Z = \text{Solar metallicity } (Z_{\odot})$ **MESA**

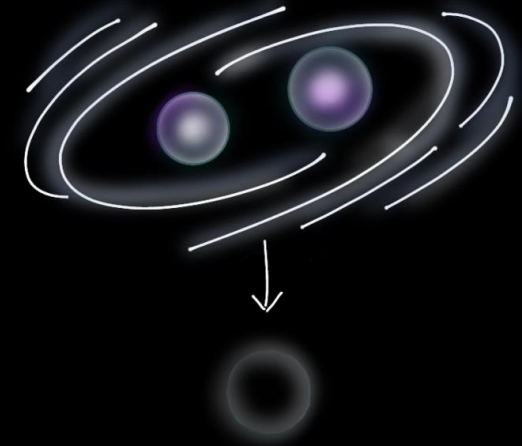
Paxton et al. 2011

*Project I, Part A***Simulation of Double Neutron Stars****Modelling supernova and kick physics**

Mandel & Müller 2020,
Kapil et al. 2023

**Post-Supernova effects on orbital dynamics**

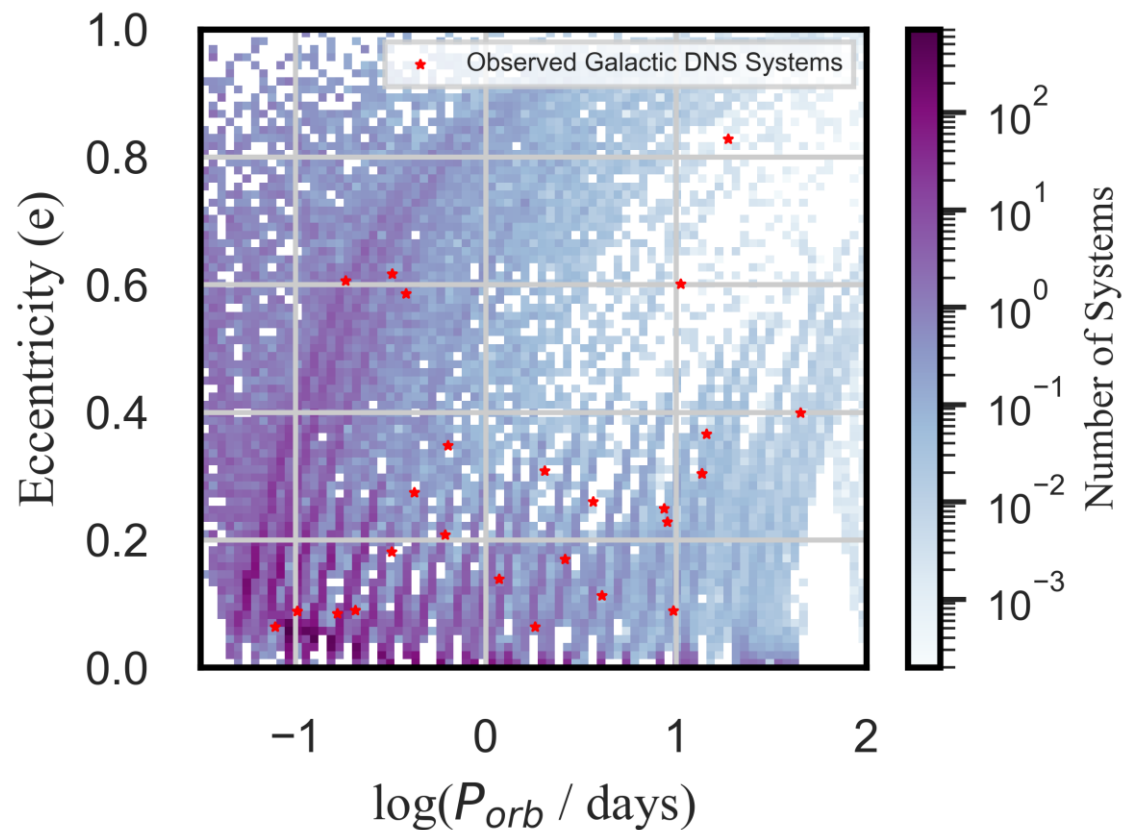
Brandt &
Podsiadlowski 1995

**Gravitational-wave merger timescale**

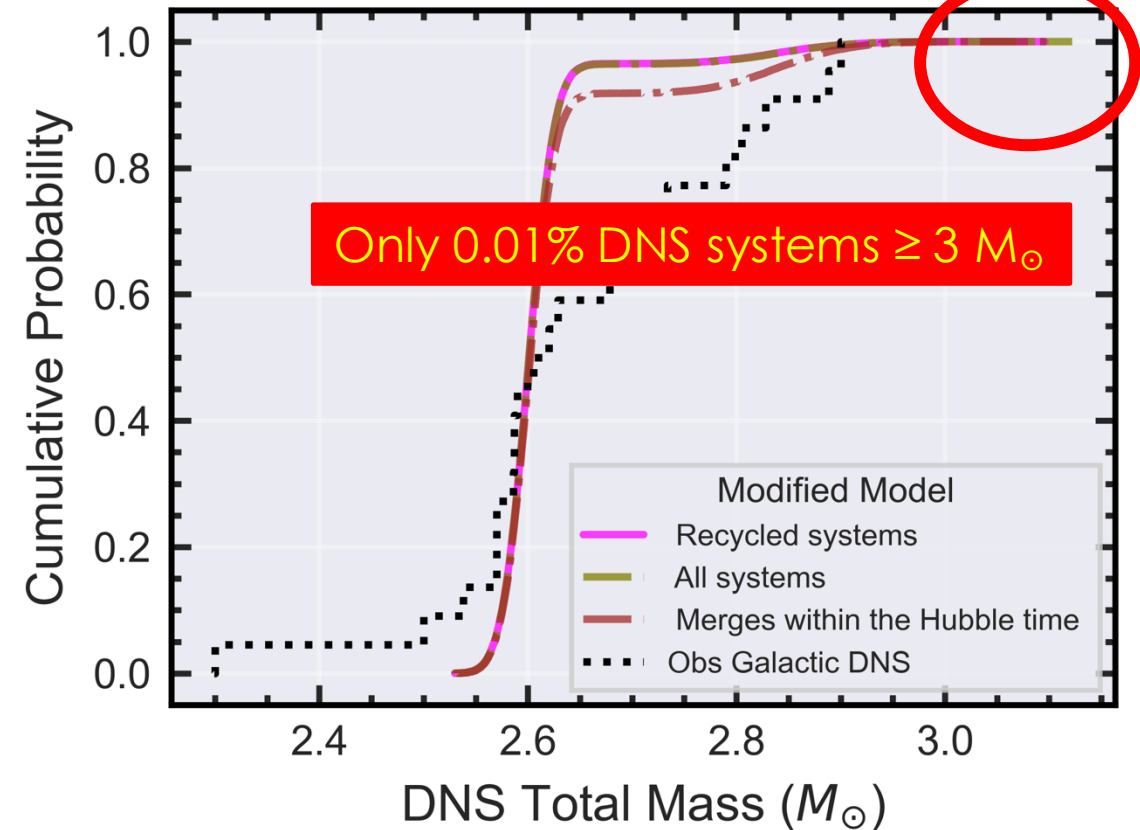
Peters 1964b
Mandel (2021b)

Simulated DNS population compared against observed DNS population ($NS1=1.4 M_{\odot}, Z = Z_{\odot}$)

Combined Porb-e Distribution of all models



DNS Total Mass Distribution



Project I, Part B

Initial parameter space

He ZAMS : $2.5 - 10 M_{\odot}$



First-born NS (NS1) : $1.1 - 1.9 M_{\odot}$

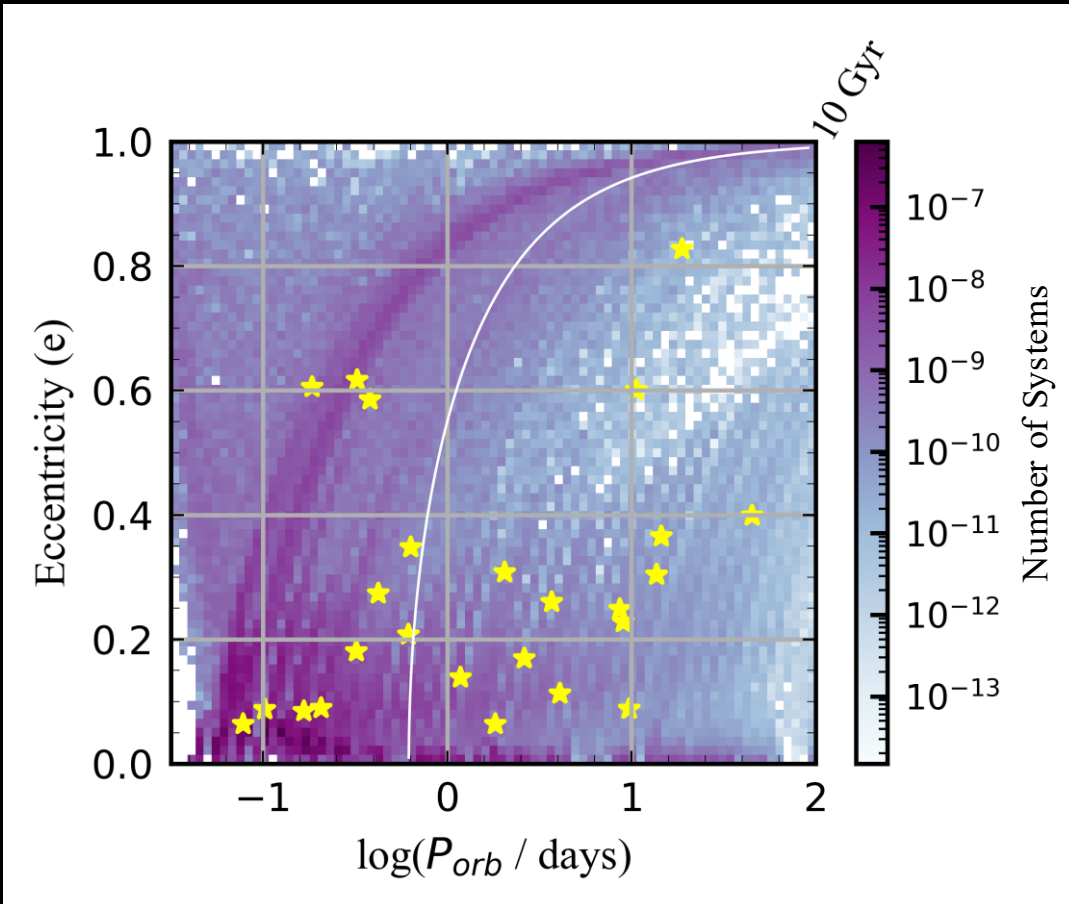
Orbital period : 1.5 hrs – 100 days



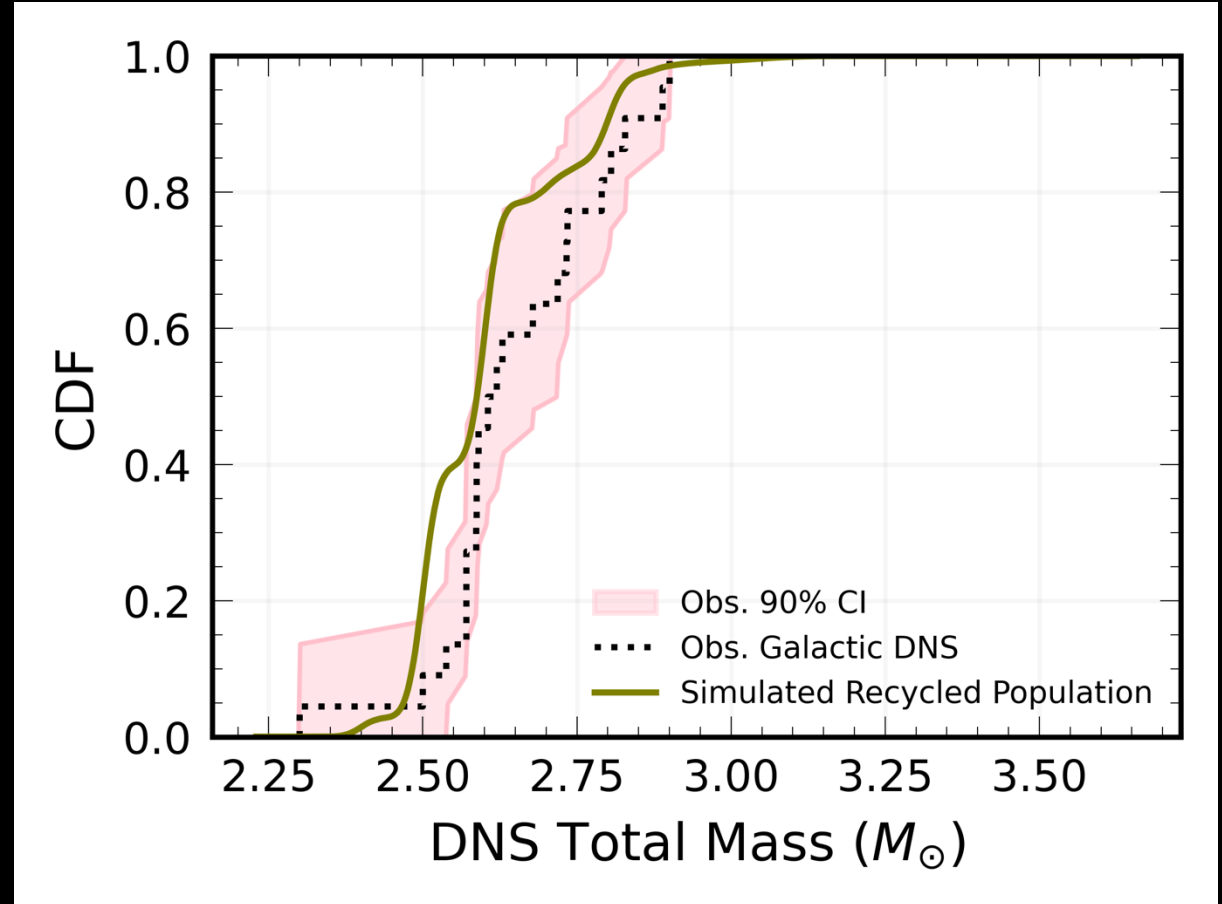
Metallicity : $Z = Z_{\odot}$, $Z = 0.1 Z_{\odot}$ and $Z = 0.01 Z_{\odot}$

Simulated DNS population compared against observed DNS population ($Z = Z_{\odot}$)

Combined Porb-e Distribution of all models



DNS Total Mass Distribution



Formation of Double Neutron Stars at Solar Metallicity

GW190425 Progenitors

a) Standard channel

$$\frac{\text{DNS} \geq 3 M_{\odot}}{\text{Total He} - \text{NS binaries}}$$

0.47%

We have a winner!

b) Massive He star

0.00006%

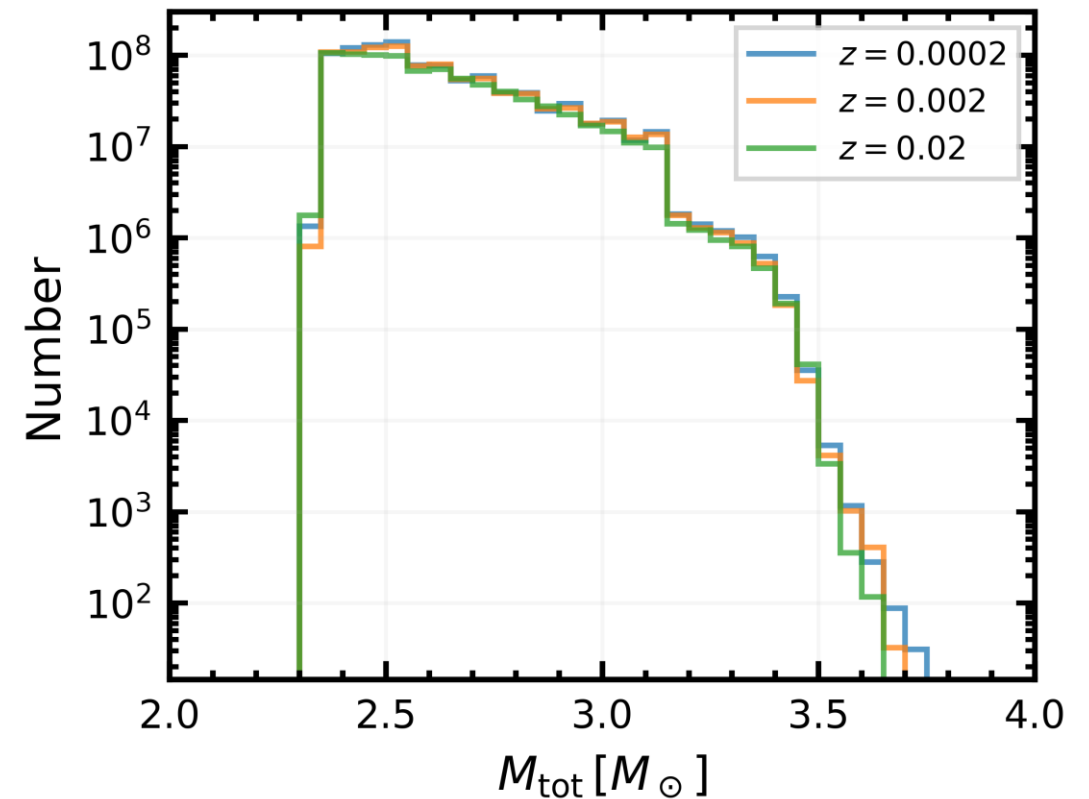
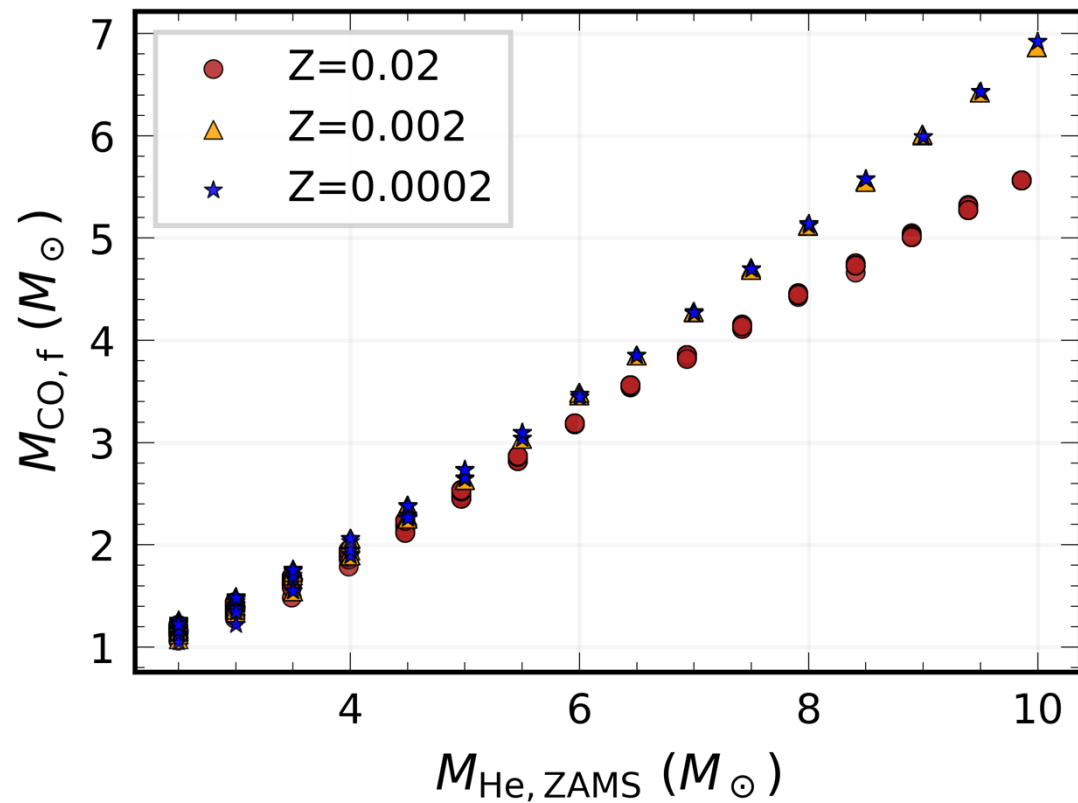
c) Massive NS1

0.0002%

Massive He stars and Massive NS1 are rare at solar metallicity ($Z_{\odot}$)!

Formation of Double Neutron Stars at Low Metallicity

GW190425 Progenitors



1. Choice of the remnant mass prescription that maps core mass to remnant mass
2. Maybe metallicity has a stronger impact during the initial evolutionary phases?
3. The parameter space that become He-NS binaries that decide the final fate of the DNS total mass might vary.

Work done so far, summarized

- Calibrated models to explain the Galactic DNS population
- Expanded the initial parameter space
- Tested the formation channels to identify dominant GW190425 progenitors
- Standard formation pathway dominates the formation of GW190425 like heavy systems at $Z_{\odot}$
- Rules out formation through unstable mass transfer
- Massive DNS formation weakly affected by metallicity

Takeaways

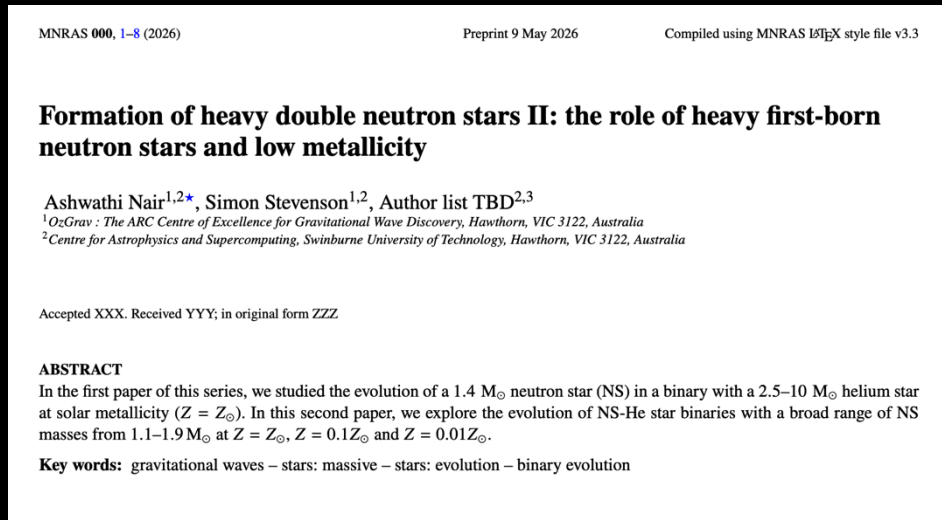
Maybe we don't need a new formation channel to explain the formation of GW190425 at $Z_{\odot}$

We see a weak dependence of metallicity on the formation of heavy systems (only 0.5% more systems)

Understanding mass distribution of neutron stars is important

Future Work

Next six months



- Formation of heavy DNS at super-Eddington
- Formation of heavy NSs in NSBH binaries
- Write thesis

Submit thesis!

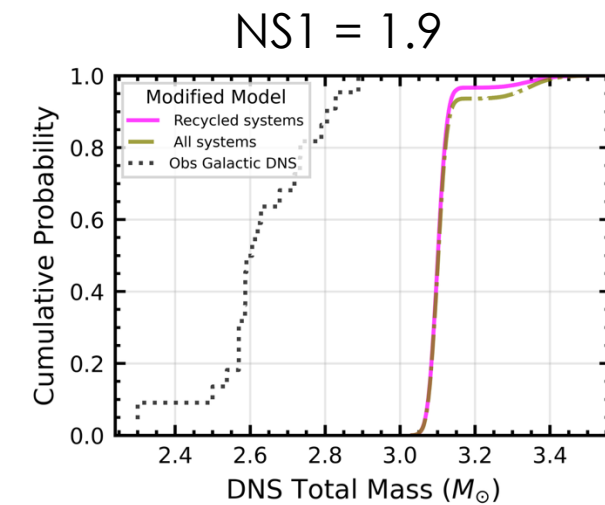
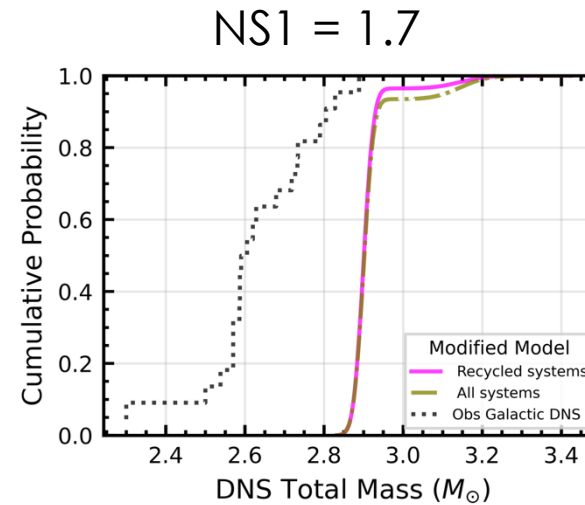
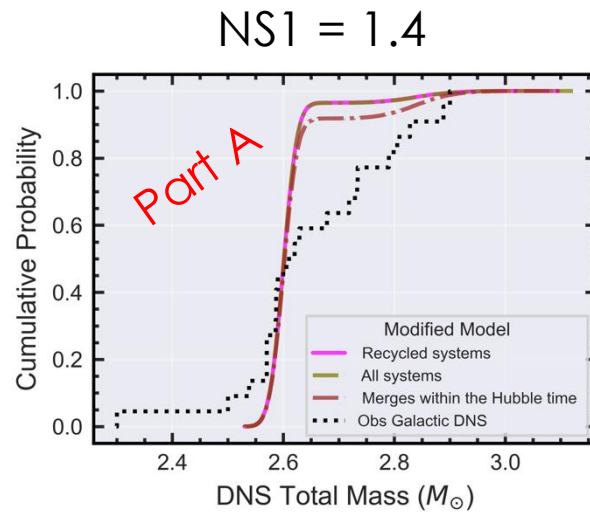
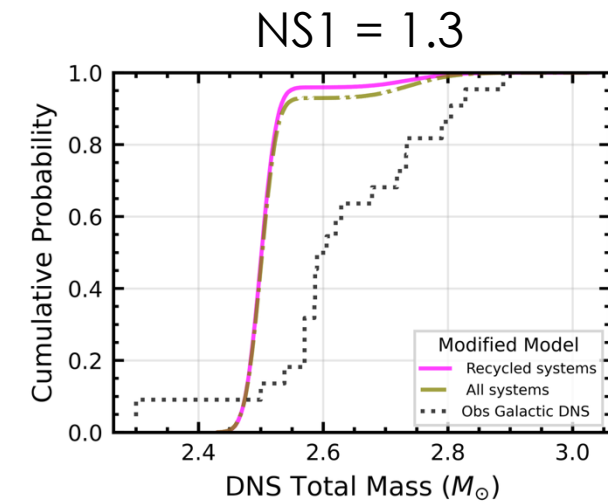
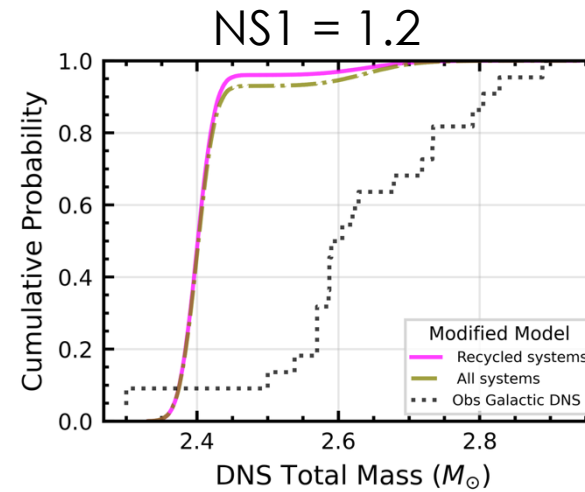
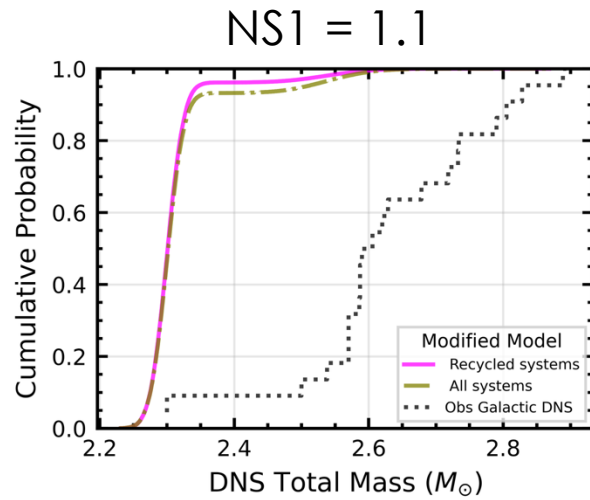
Nair et al. 2026 (in prep)

EXTRA SLIDES

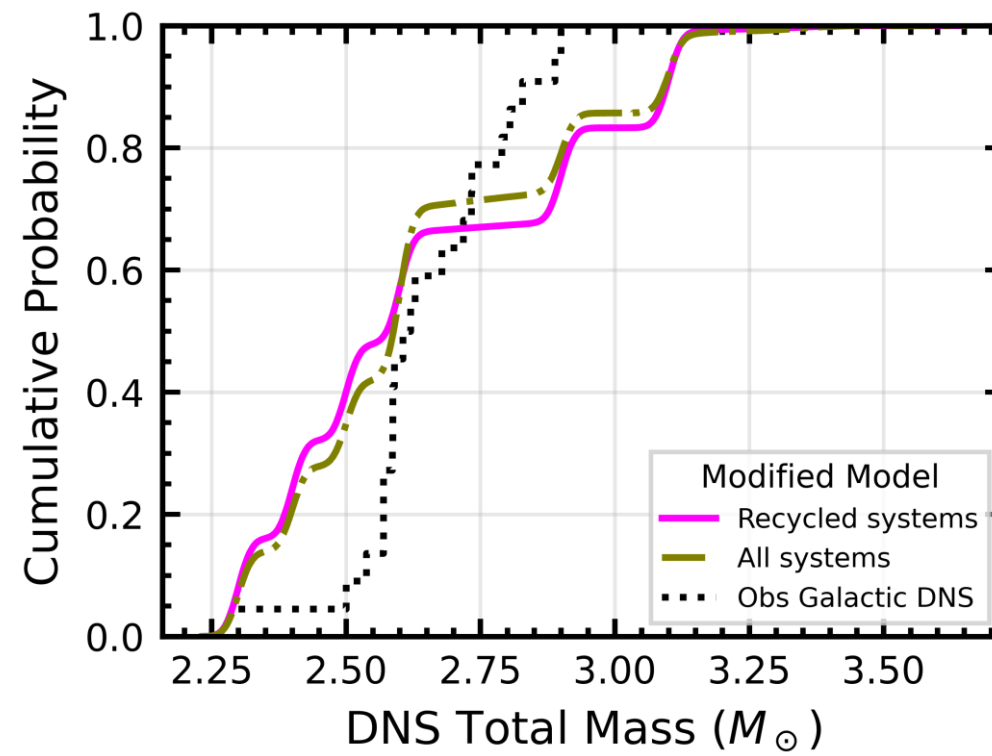
Project 1, Part B

DNS Total mass distribution

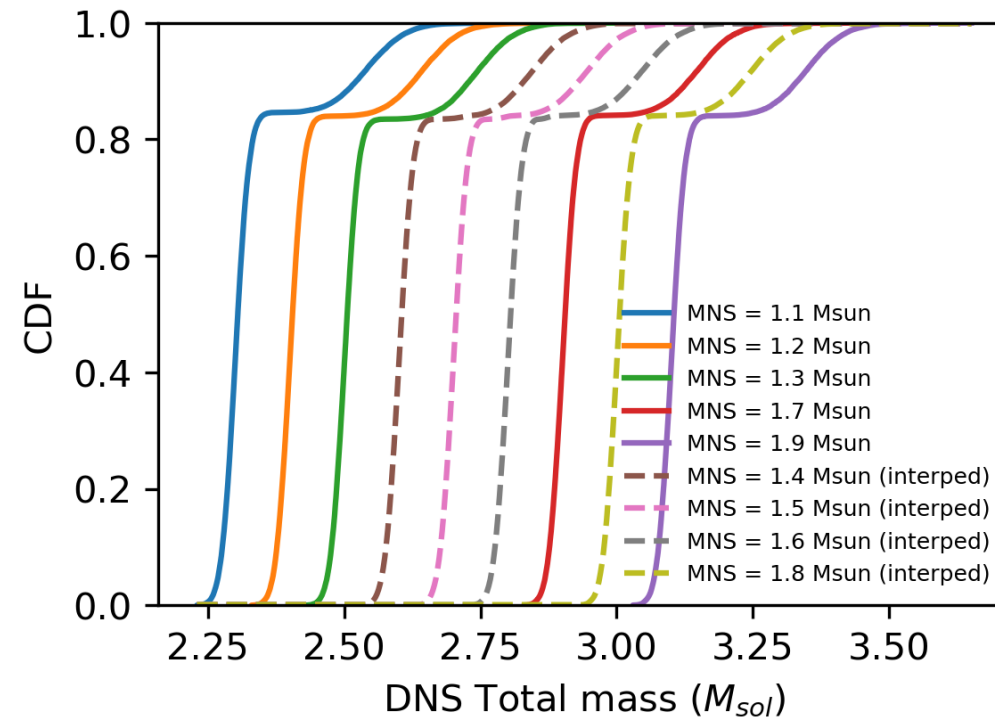
Constant NS1 (1.1, 1.2, 1.3, 1.4, 1.7, 1.9, respectively) + NS2



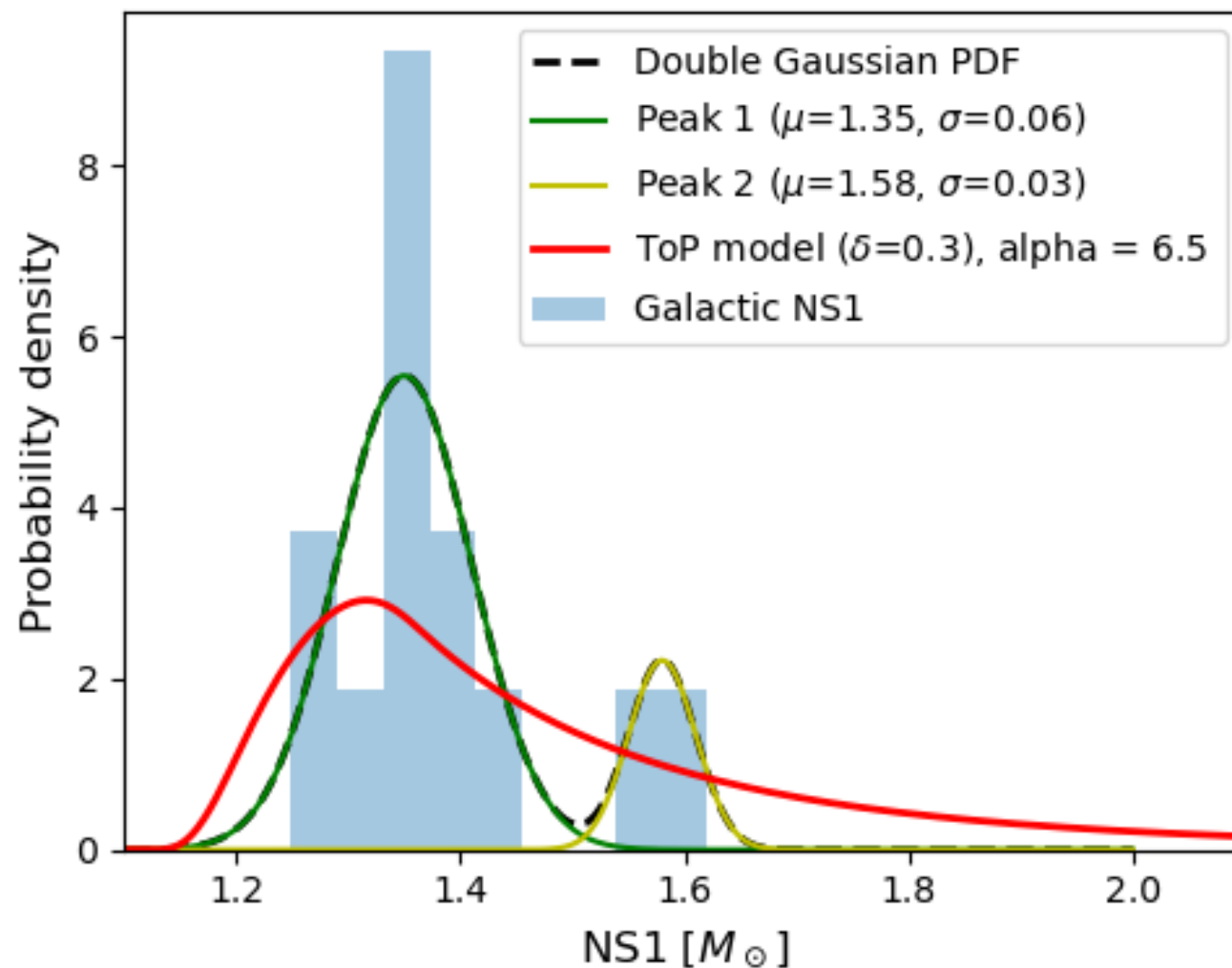
Combined CDF



Interpolated CDFs based on the simulated ones

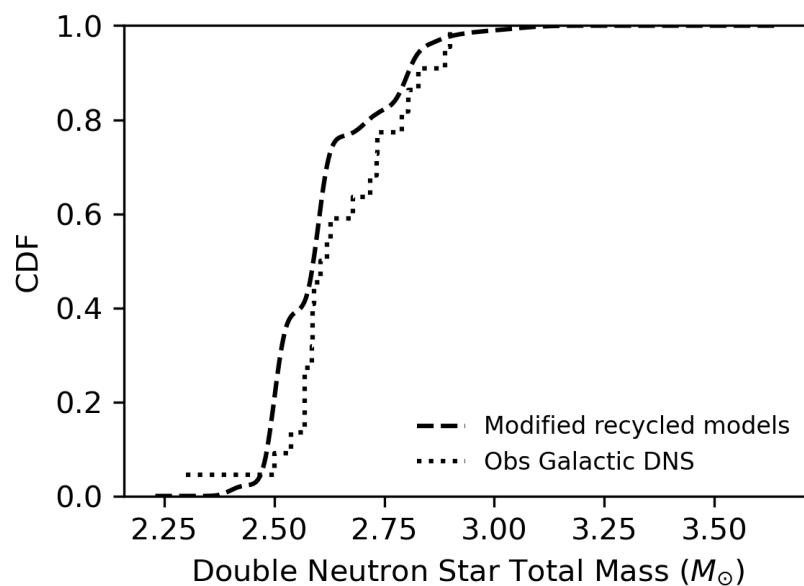


Weights from observed Galactic recycled NSs (double gaussian) and birth-mass function of neutron stars based on the turn-on power law (TOP) model from Z. You et. al 2025

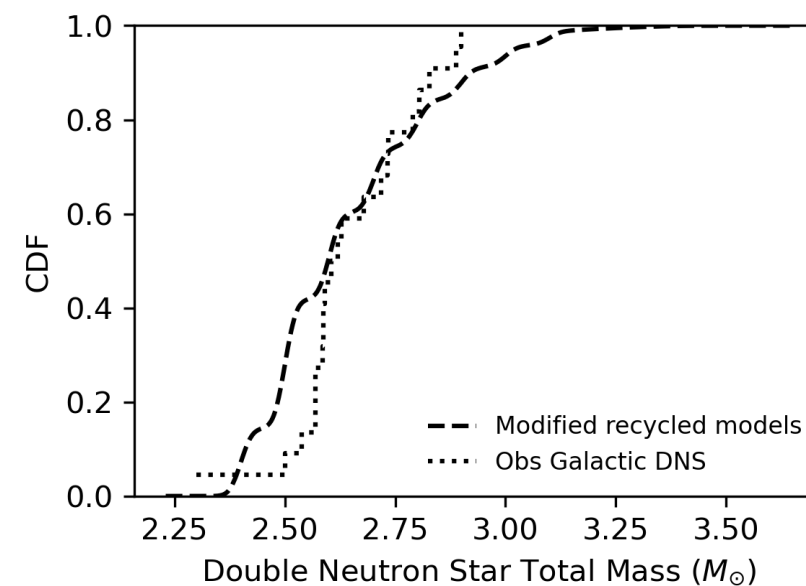


Based on weights from

Observed Galactic DNS



TOP Model
Z. You et al. 2025



Recycled models (simulated) = representative of radio observed population

11/05/2026

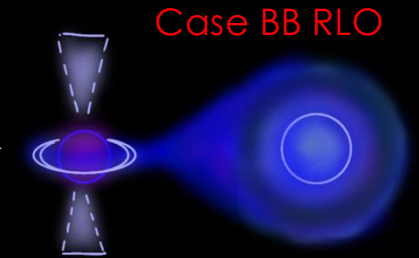
GW190425 Progenitors

Only 0.01% DNS systems $\geq 3 M_{\odot}$

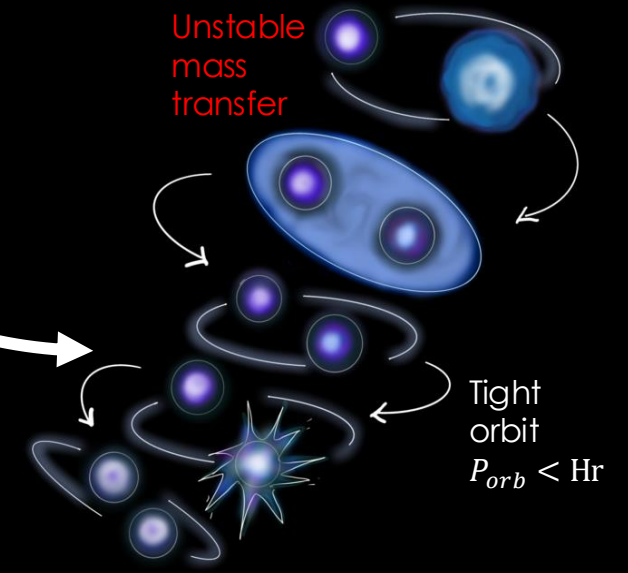


Vigna Gómez et al.
(2021)

- 82% - Standard formation pathway
- 18% - Massive helium channel
- Rules out the fast merger channel



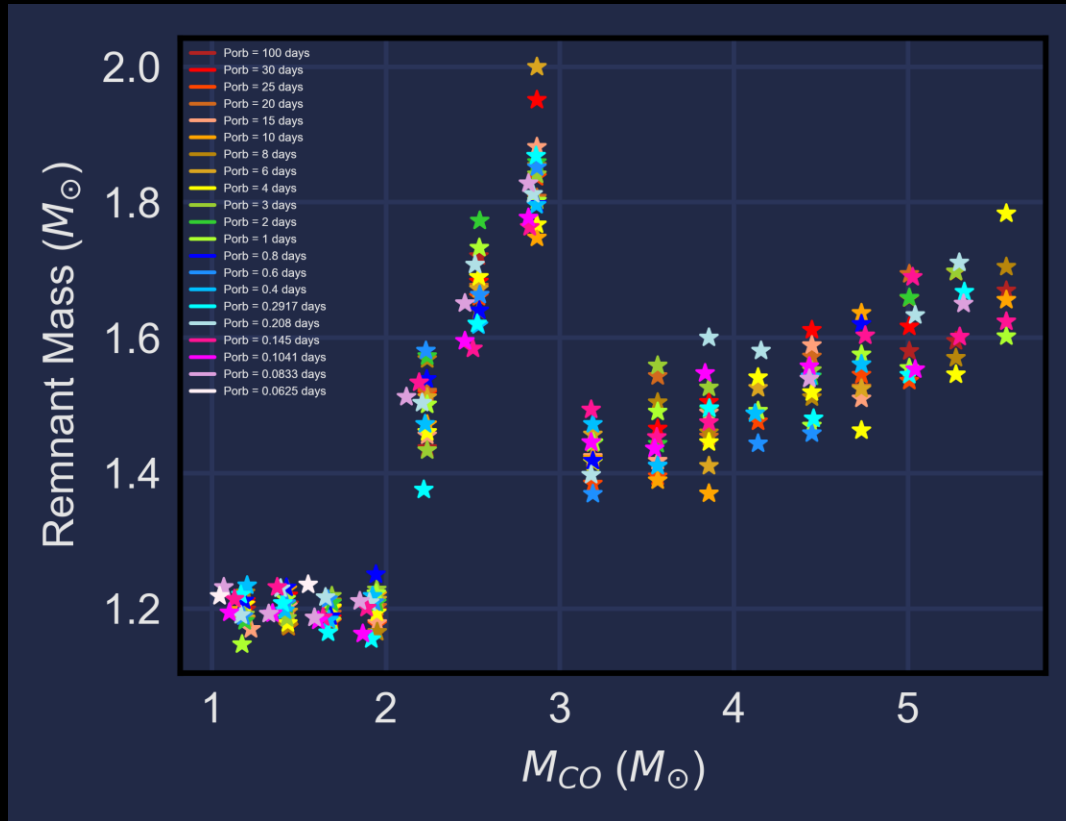
Unstable
mass
transfer



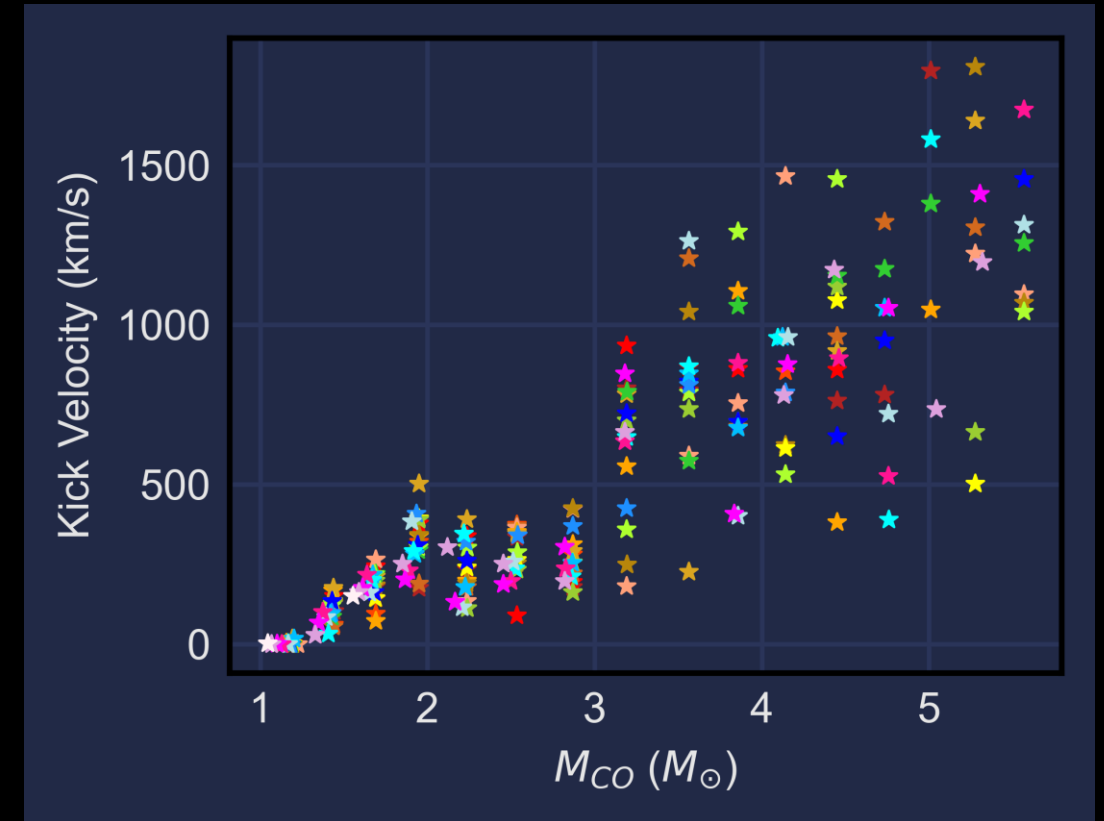
Tight orbit
 $P_{orb} < \text{Hr}$

B.P. Abbott et al. (2020)

Standard Supernovae and Kick Prescription (Standard Model)



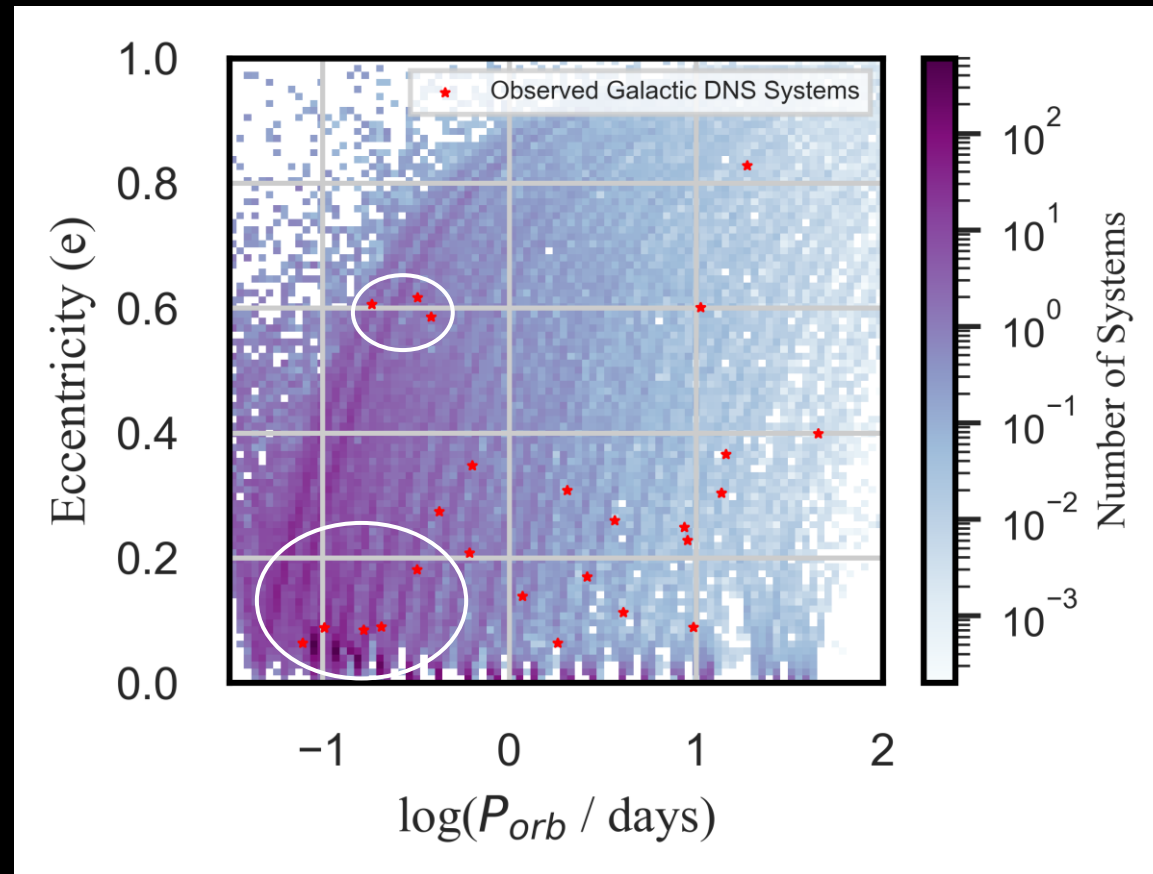
Remnant masses (NS2) as a function of CO core mass



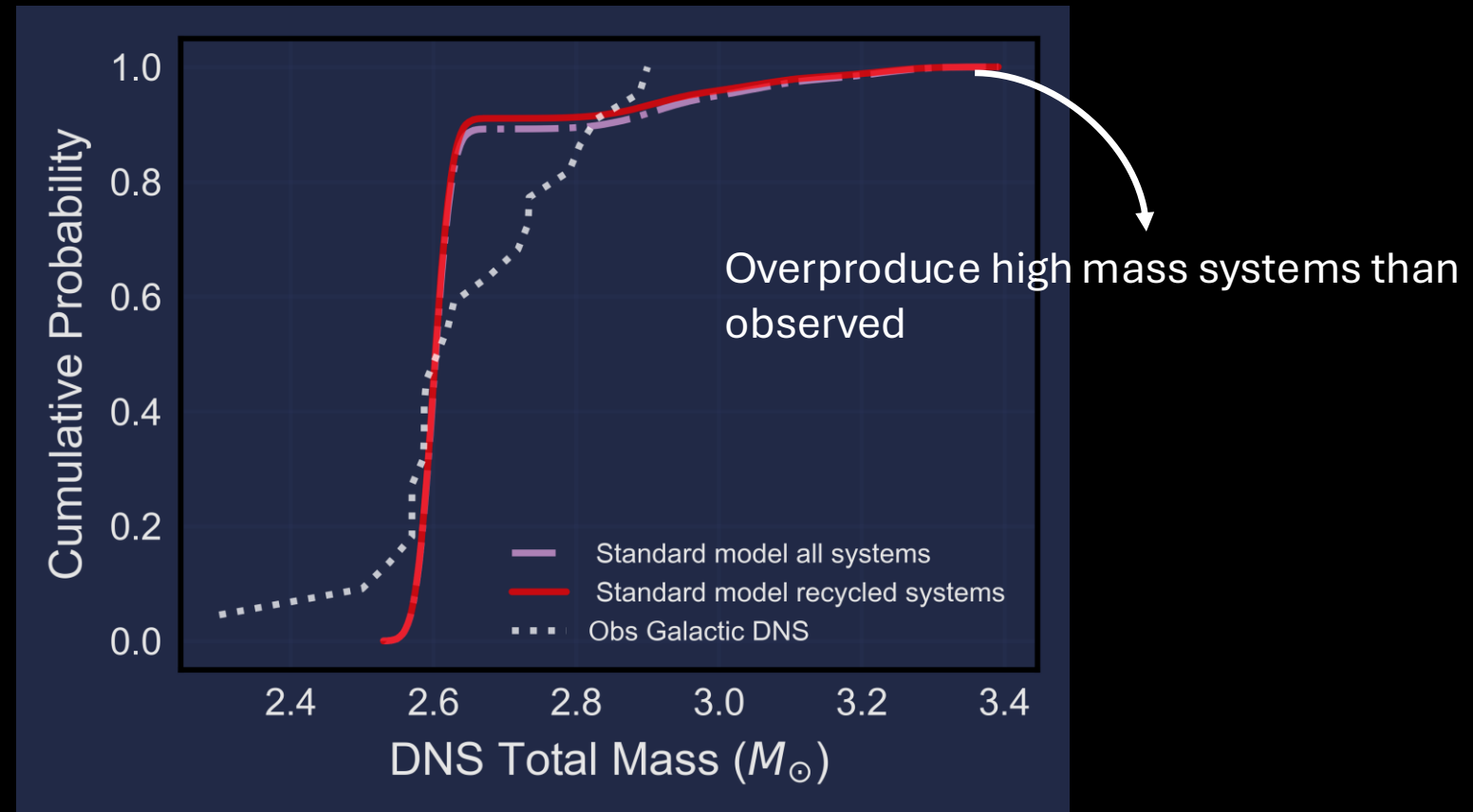
Kick velocity for the corresponding NS2 as a function of CO core mass

Combined P_{orb} -e Distribution of all models

Standard Model



DNS Total Mass Distribution Standard Model



Issues with the Standard Prescription:

- Overpredicts eccentricities
- Overpredicts DNS total mass within the recycled systems

Modifications to the Standard Model

Why?

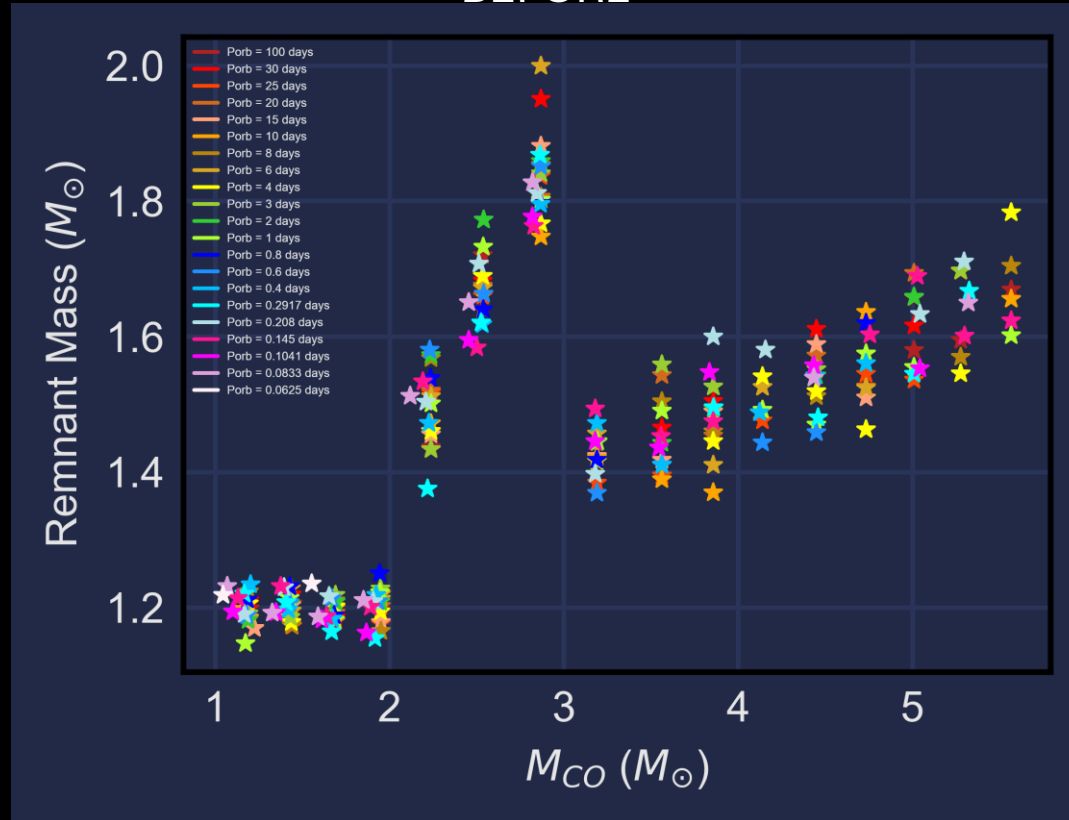
To calibrate my models to explain the observed galactic DNS population

What?

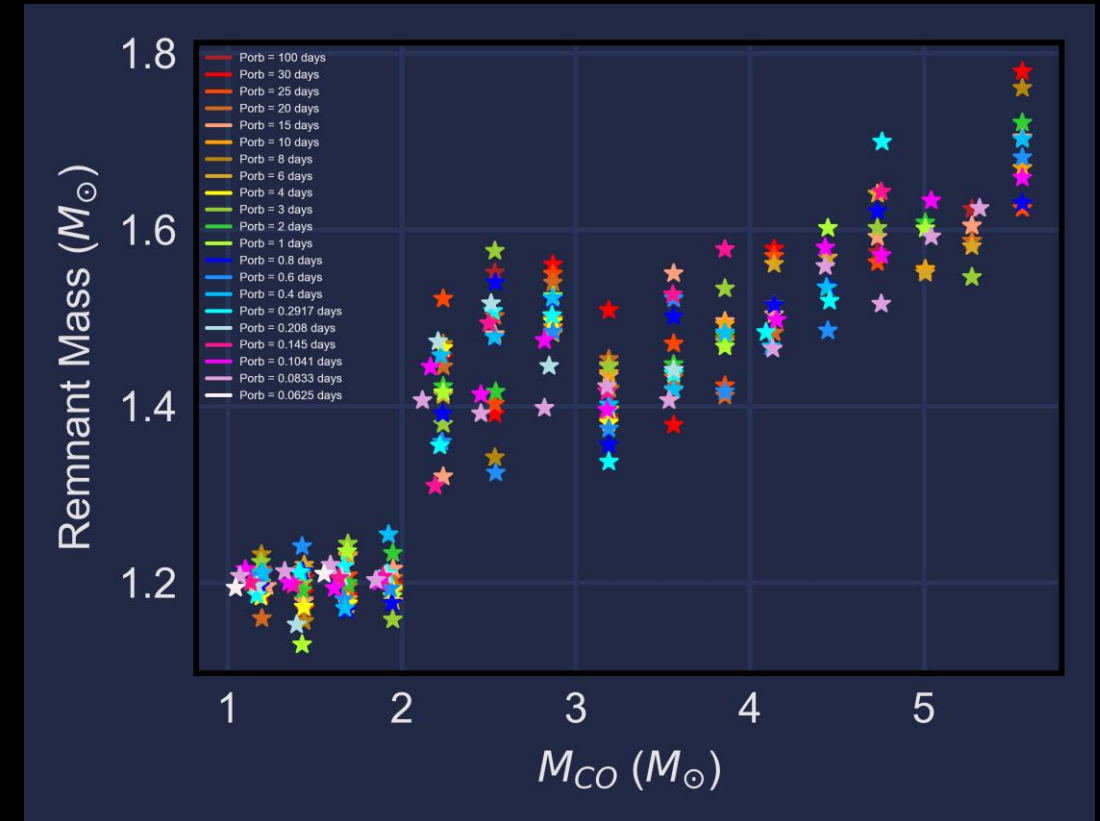
1. The NS scaling factor
2. Defined USSN and ECSN
3. High velocity kicks to slow merging systems

Standard Model

BEFORE



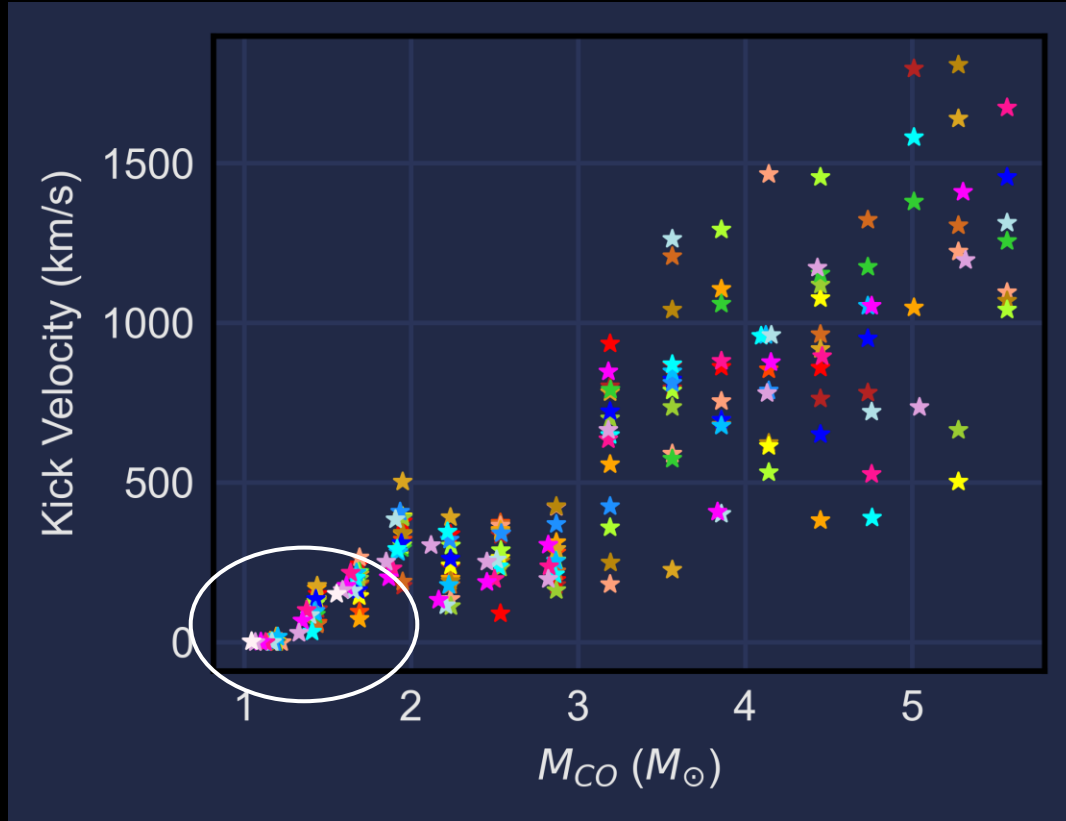
Modified Model



Remnant masses (NS2) as a function of CO core mass

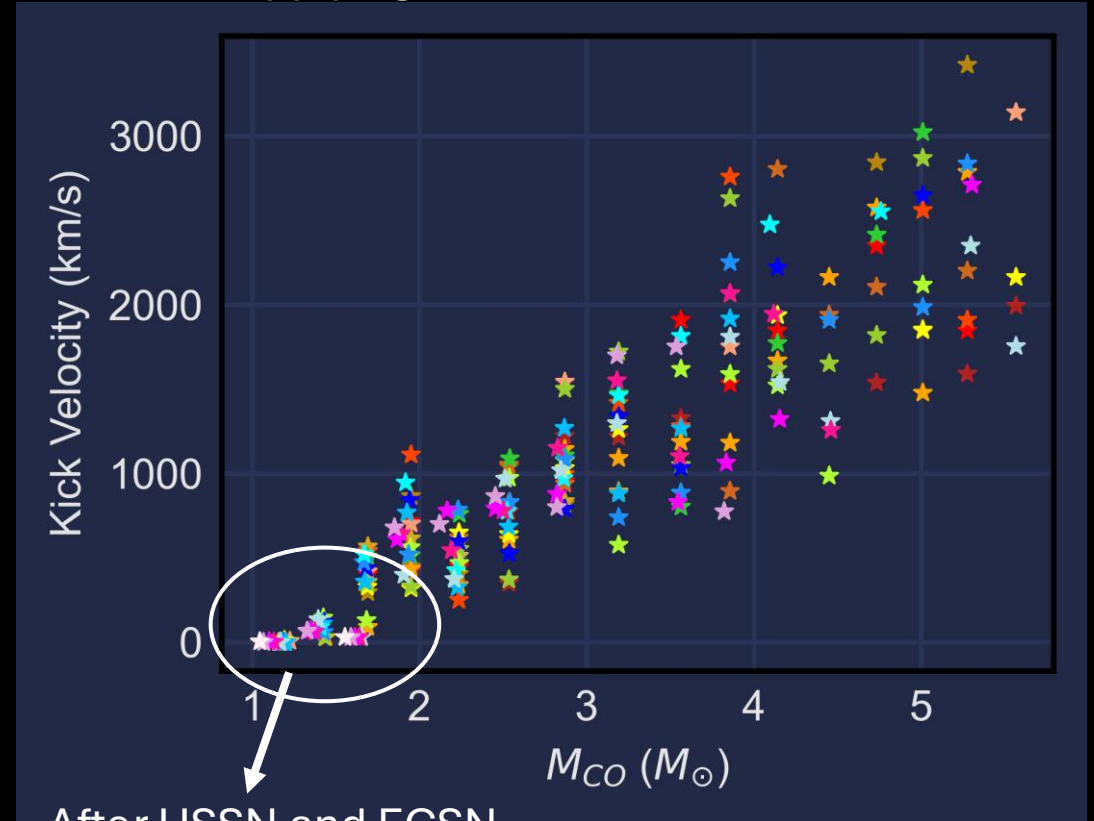
Standard Model

Before applying USSN and ECSN condition



Modified Model

After applying USSN and ECSN condition



Kick velocity for the corresponding NS2 as a function of CO core mass